

Program for ESP8266:

```
import cv2
import numpy as np
from ultralytics import YOLO
import threading
from kiwislot import Kiwislot
import time

# Load YOLOv8n model
model = YOLO("yolov8n.pt")

# Kiwislot Setup
iot = Kiwislot(topic="dash_1774438910648")
iot.connect()
iot.loop_start()

# ESP32-CAM URLs
urls = [
    "http://192.168.137.191:81/stream",
    "http://192.168.137.187:81/stream",
    "http://192.168.137.157:81/stream",
    "http://192.168.137.137:81/stream"
]

width, height = 320, 240

class CamThread:
    def __init__(self, url, name):
        self.cap = cv2.VideoCapture(url)
        self.frame = np.zeros((height, width, 3), dtype=np.uint8)
        self.ret = False
        self.name = name
        self.running = True

        self.thread = threading.Thread(target=self.update, daemon=True)
        self.thread.start()
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def update(self):
    while self.running:
        ret, frame = self.cap.read()
        if ret:
            frame = cv2.resize(frame, (width, height))
            self.frame = frame
            self.ret = True
        else:
            self.ret = False

def read(self):
    return self.ret, self.frame.copy()

def stop(self):
    self.running = False
    self.cap.release()

# Initialize Cameras
cams = [CamThread(urls[i], f"CAM {i+1}") for i in range(4)]

prev_state = None

while True:
    frames = []
    person_detected_any = False

    for cam in cams:
        ret, frame = cam.read()

        if not ret:
            frame = np.zeros((height, width, 3), dtype=np.uint8)
            cv2.putText(frame, f"{cam.name} Offline", (40, 120),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 0, 255), 2)
        else:

            # YOLOv8 Detection

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results = model(frame, verbose=False)

for r in results:
    for box in r.bboxes:
        cls_id = int(box.cls[0])
        conf = float(box.conf[0])

        if cls_id == 0 and conf > 0.5:
            person_detected_any = True

            x1, y1, x2, y2 = map(int, box.xyxy[0])

            cv2.rectangle(frame, (x1, y1), (x2, y2),
                          (0, 255, 0), 2)
            cv2.putText(frame, f"Person {conf:.2f}",
                        (x1, y1 - 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.6,
                        (0, 255, 0), 2)

cv2.putText(frame, cam.name, (10, 30),
            cv2.FONT_HERSHEY_SIMPLEX, 0.8, (255, 255, 0), 2)

frames.append(frame)

# Send ONLY when state changes
if prev_state is None:
    prev_state = person_detected_any

if person_detected_any != prev_state:
    if person_detected_any:
        print(" 📡 Sending 1 (Person Detected)")
        iot.send("3", "1")
    else:
        print(" 📡 Sending 0 (No Person)")
        iot.send("3", "0")

prev_state = person_detected_any

```

```

# Combine 2x2 grid
top = np.hstack((frames[0], frames[1]))
bottom = np.hstack((frames[2], frames[3]))
combined = np.vstack((top, bottom))

cv2.imshow("YOLOv8 Multi-Cam (Cloud)", combined)

if cv2.waitKey(1) & 0xFF == ord('q'):
    break

# Cleanup
for cam in cams:
    cam.stop()

cv2.destroyAllWindows()
#include <KiwisIoT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#define MAP_CH "1"
// ===== WIFI =====
const char* ssid = "error";
const char* pass = "12345678";
const char* topic = "dash_1774438910648";

float lat = 10.929296339763807;
float lng = 78.7381934818362;

KiwisIoT kiwisiot(ssid, pass, topic);

// ===== MOTOR PINS =====
#define IN1 D3
#define IN2 D4
#define IN3 D5
#define IN4 D6

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#define LIGHT_PIN D1
#define HORN_PIN D2

// ===== SENSORS =====
#define TRIG_PIN D7
#define ECHO_PIN D8
#define IR_PIN D0

// ===== LCD =====
LiquidCrystal_I2C lcd(0x27, 16, 2);

int powerState = 0;

// Sensor variables
long duration;
int distance;
int irValue;

// SMS control
bool smsSent = false;

// ----- GSM FUNCTION -----
void sendSMS(String msg) {
  Serial.println("AT");
  delay(1000);

  Serial.println("AT+CMGF=1");
  delay(1000);

  Serial.println("AT+CMGS=\"" + 916374593717 + "\""); // PUT YOUR
NUMBER
  delay(1000);

  Serial.print(msg);
  delay(500);

```

```
Serial.write(26); // CTRL+Z
delay(5000);
}
```

```
// ----- MOTOR CONTROL -----
```

```
void motorForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
```

```
void motorReverse() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}
```

```
void motorRight() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}
```

```
void motorLeft() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
```

```
void motorStop() {
  digitalWrite(IN1, LOW);
}
```

```

digitalWrite(IN2, LOW);
digitalWrite(IN3, LOW);
digitalWrite(IN4, LOW);
}

// ----- SENSOR FUNCTION -----

void readSensors() {

  // Ultrasonic
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH);
  distance = duration * 0.034 / 2;

  // IR
  irValue = digitalRead(IR_PIN);

  // Serial output
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm | IR: ");
  Serial.print(irValue == 0 ? "ON" : "OFF");
  Serial.print(" | Status: ");

  // ----- CONDITION -----
  if (distance < 10 || irValue == 0) {

    motorStop();
    Serial.println("CAR STOPPED");

    // 🚩 SEND SMS ONLY ONCE

```

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    if (!smsSent) {
        String link = "https://maps.google.com/?q=" + String(lat,6) + "," +
String(lng,6);
        sendSMS("Obstacle Detected!\nLocation: " + link);
        smsSent = true;
    }

    lcd.setCursor(0, 0);
    lcd.print("CAR STOPPED ");

} else {

    Serial.println("RUNNING");
    smsSent = false;

    lcd.setCursor(0, 0);
    lcd.print("CAR RUNNING ");
}

// ----- LCD Line 2 -----
lcd.setCursor(0, 1);
lcd.print("D:");
lcd.print(distance);
lcd.print(" IR:");
lcd.print(irValue == 0 ? "ON " : "OFF");
}

// ----- IOT CONTROL -----

void onControl(String ch, String val) {

    Serial.print("CH: ");
    Serial.print(ch);
    Serial.print(" | VAL: ");
    Serial.println(val);

```

```

// ● PRIORITY: CHANNEL 3 EMERGENCY STOP
if (ch == "3") {
  if (val == "1") {
    motorStop();
    powerState = 0; // optional: also turn off power
    Serial.println(" 🚨 EMERGENCY STOP (CH3)");
    String link = "https://maps.google.com/?q=" + String(lat,6) + "," +
String(lng,6);
    sendSMS("Manual Request\nLocation: " + link);

    lcd.setCursor(0, 0);
    lcd.print("EMERGENCY STOP ");

    return; // STOP everything else
  }
}

// ----- NORMAL CONTROL -----
if (ch != "0") return;

if (val == "power:1") {
  powerState = 1;
  Serial.println("Power ON");
  return;
}

if (val == "power:0") {
  powerState = 0;
  motorStop();
  Serial.println("Power OFF");
  return;
}

if (val == "lights:1") {
  digitalWrite(LIGHT_PIN, HIGH);
  return;
}

```

```
if (val == "lights:0") {  
    digitalWrite(LIGHT_PIN, LOW);  
    return;  
}
```

```
if (val == "horn") {  
    digitalWrite(HORN_PIN, HIGH);  
    delay(300);  
    digitalWrite(HORN_PIN, LOW);  
    return;  
}
```

```
if (powerState == 0) return;
```

```
if (val == "forward") motorForward();  
else if (val == "back") motorReverse();  
else if (val == "right") motorRight();  
else if (val == "left") motorLeft();  
else if (val == "stop") motorStop();  
}
```

```
// ----- SETUP -----
```

```
void setup() {
```

```
    Serial.begin(9600); // GSM uses Serial
```

```
    pinMode(IN1, OUTPUT);  
    pinMode(IN2, OUTPUT);  
    pinMode(IN3, OUTPUT);  
    pinMode(IN4, OUTPUT);
```

```
    pinMode(LIGHT_PIN, OUTPUT);  
    pinMode(HORN_PIN, OUTPUT);
```

```

pinMode(TRIG_PIN, OUTPUT);
pinMode(ECHO_PIN, INPUT);
pinMode(IR_PIN, INPUT);

motorStop();

// LCD
lcd.init();
lcd.backlight();
lcd.setCursor(0, 0);
lcd.print("System Ready");
delay(1000);
lcd.clear();

kiwisiot.begin();
kiwisiot.onReceive(onControl);
}

// ----- LOOP -----

void loop() {
    kiwisiot.run();

    // 📧 Serial command to send location
    if (Serial.available()) {
        char c = Serial.read();

        if (c == 'l') {
            String link = "https://maps.google.com/?q=" + String(lat,6) + "," +
String(lng,6);
            sendSMS("Manual Request\nLocation: " + link);
        }
    }

    static unsigned long lastSend = 0;

```

```
if (millis() - lastSend > 3000) {  
    lastSend = millis();  
  
    Serial.print("LAT: ");  
    Serial.println(lat, 6);  
    Serial.print("LNG: ");  
    Serial.println(lng, 6);  
  
    // Convert to string format "lat,lng"  
    String loc = String(lat, 6) + "," + String(lng, 6);  
  
    // Send to map channel  
    kiwisiot.send(MAP_CH, loc);  
}  
readSensors();  
delay(500);  
}
```